

DDR4-UDIMM component guide

Every component by designator, with the net on each pin, what that net connects to, and the layers it may use. Companion to ROUTING_GUIDE.pdf, which is organised by layer. Read from DDR4-UDIMM — 16 GB DDR4.PcbDoc.

1. What is on the board

Prefix	What	Count
J1	288-pin DDR4 UDIMM edge connector (5 schematic parts, one footprint)	1
U1-U8	Micron MT40A2G8SA-062E:F DRAM, 78-ball FBGA	8
U9	Microchip 34AA04 SPD EEPROM	1
R1-R126	series, termination and pull-up resistors	126
C1-C70	decoupling and bulk capacitors	70

Power and ground pins are left out of the per-pin tables below - they are connected to the planes with their own vias, not routed as signals.

2. J1 - edge connector

Signal pins only. Every one of these runs to a resistor first, never straight to a DRAM.

J1 pin	Net	Class	Goes to	Layers	copper mm
3	DQ4	Data	R6-2	L1, L3, L8	2.95
5	DQ0	Data	R2-2	L1, L3, L8	2.95
7	DM0	Data	R11-2	L1, L3, L8	2.95
10	DQ6	Data	R7-2	L1, L3, L8	2.95
12	DQ2	Data	R4-2	L1, L3, L8	2.95
14	DQ12	Data	R18-2	L1, L3, L8	2.95
16	DQ8	Data	R14-2	L1, L3, L8	2.95
18	DM1	Data	R23-2	L1, L3, L8	2.95
21	DQ14	Data	R19-2	L1, L3, L8	2.95
23	DQ10	Data	R16-2	L1, L3, L8	2.95
25	DQ20	Data	R30-2	L1, L3, L8	2.95
27	DQ16	Data	R26-2	L1, L3, L8	2.95
29	DM2	Data	R35-2	L1, L3, L8	2.95
32	DQ22	Data	R31-2	L1, L3, L8	2.95
34	DQ18	Data	R28-2	L1, L3, L8	2.95
36	DQ28	Data	R42-2	L1, L3, L8	2.97
38	DQ24	Data	R38-2	L1, L3, L8	3.01
40	DM3	Data	R47-2	L1, L3, L8	3.05
43	DQ30	Data	R43-2	L1, L3, L8	3.12
45	DQ26	Data	R40-2	L1, L3, L8	3.16
58	RESET_N	Control	U1-L1, U2-L1, U3-L1, U4-L1, U5-L1, U6-L1, U7-L1, U8-L1	L1, L3, L5, L6, L8	4.53
60	CKE	Control	R104-1, U1-G3, U2-G3, U3-G3, U4-G3, U5-G3, U6-G3, U7-G3, U8-G3	L1, L3, L5, L6, L8	4.53
62	ACT	Address	R125-1, U1-H3, U2-H3, U3-H3, U4-H3, U5-H3, U6-H3, U7-H3, U8-H3	L1, L3, L5, L6, L8	4.53
63	BG0	Address	R99-1, U1-J2, U2-J2, U3-J2, U4-J2, U5-J2, U6-J2, U7-J2, U8-J2	L1, L3, L5, L6, L8	4.53
65	A12_BC	Address	R123-1, U1-J7, U2-J7, U3-J7, U4-J7, U5-J7, U6-J7, U7-J7, U8-J7	L1, L3, L5, L6, L8	4.53
66	A9	Address	R119-1, U1-M7, U2-M7, U3-M7, U4-M7, U5-M7, U6-M7, U7-M7, U8-M7	L1, L3, L5, L6, L8	4.53
68	A8	Address	R116-1, U1-M2, U2-M2, U3-M2, U4-M2, U5-M2, U6-M2, U7-M2, U8-M2	L1, L3, L5, L6, L8	4.53
69	A6	Address	R112-1, U1-L2, U2-L2, U3-L2, U4-L2, U5-L2, U6-L2, U7-L2, U8-L2	L1, L3, L5, L6, L8	4.53

71	A3	Address	R105-1, U1-K7, U2-K7, U3-K7, U4-K7, U5-K7, U6-K7, U7-K7, U8-K7	L1, L3, L5, L6, L8	4.53
72	A1	Address	R100-1, U1-L7, U2-L7, U3-L7, U4-L7, U5-L7, U6-L7, U7-L7, U8-L7	L1, L3, L5, L6, L8	4.53
74	CK0_T	Clock	R115-1, U1-F7, U2-F7, U3-F7, U4-F7, U5-F7, U6-F7, U7-F7, U8-F7	L1, L6, L8	4.53
75	CK0_C	Clock	R117-1, U1-F8, U2-F8, U3-F8, U4-F8, U5-F8, U6-F8, U7-F8, U8-F8	L1, L6, L8	4.53
79	A0	Address	R97-1, U1-L3, U2-L3, U3-L3, U4-L3, U5-L3, U6-L3, U7-L3, U8-L3	L1, L3, L5, L6, L8	4.53
81	BA0	Address	R118-1, U1-K2, U2-K2, U3-K2, U4-K2, U5-K2, U6-K2, U7-K2, U8-K2	L1, L3, L5, L6, L8	4.53
82	A16_RAS	Address	R113-1, U1-H8, U2-H8, U3-H8, U4-H8, U5-H8, U6-H8, U7-H8, U8-H8	L1, L3, L5, L6, L8	4.53
84	CS	Control	R101-1, U1-G7, U2-G7, U3-G7, U4-G7, U5-G7, U6-G7, U7-G7, U8-G7	L1, L3, L5, L6, L8	4.53
86	A15_CAS	Address	R111-1, U1-H7, U2-H7, U3-H7, U4-H7, U5-H7, U6-H7, U7-H7, U8-H7	L1, L3, L5, L6, L8	4.53
87	ODT	Control	R107-1, U1-F3, U2-F3, U3-F3, U4-F3, U5-F3, U6-F3, U7-F3, U8-F3	L1, L3, L5, L6, L8	4.53
95	DQ36	Data	R54-2	L1, L3, L8	3.20
97	DQ32	Data	R50-2	L1, L3, L8	3.20
99	DM4	Data	R59-2	L1, L3, L8	3.20
102	DQ38	Data	R55-2	L1, L3, L8	3.20
104	DQ34	Data	R52-2	L1, L3, L8	3.20
106	DQ44	Data	R66-2	L1, L3, L8	3.18
108	DQ40	Data	R62-2	L1, L3, L8	3.14
110	DM5	Data	R71-2	L1, L3, L8	3.10
113	DQ46	Data	R67-2	L1, L3, L8	3.03
115	DQ42	Data	R64-2	L1, L3, L8	2.99
117	DQ52	Data	R78-2	L1, L3, L8	2.95
119	DQ48	Data	R74-2	L1, L3, L8	2.95
121	DM6	Data	R83-2	L1, L3, L8	2.95
124	DQ54	Data	R79-2	L1, L3, L8	2.95
126	DQ50	Data	R76-2	L1, L3, L8	2.95
128	DQ60	Data	R90-2	L1, L3, L8	2.95
130	DQ56	Data	R86-2	L1, L3, L8	2.95
132	DM7	Data	R95-2	L1, L3, L8	2.95
135	DQ62	Data	R91-2	L1, L3, L8	2.95
137	DQ58	Data	R88-2	L1, L3, L8	2.95
139	SA0	SPD	U9-1	any signal layer	0.00
140	SA1	SPD	U9-2	any signal layer	0.00
141	SCL	SPD	U9-6	any signal layer	0.00
148	DQ5	Data	R12-2	L1, L3, L8	5.45
150	DQ1	Data	R3-2	L1, L3, L8	5.45
152	DQS0_C	Data	R10-2	L1, L3, L8	5.54
153	DQS0_T	Data	R9-2	L1, L3, L8	5.54
155	DQ7	Data	R8-2	L1, L3, L8	5.45
157	DQ3	Data	R5-2	L1, L3, L8	5.45
159	DQ13	Data	R24-2	L1, L3, L8	5.45
161	DQ9	Data	R15-2	L1, L3, L8	5.47
163	DQS1_C	Data	R22-2	L1, L3, L8	5.54
164	DQS1_T	Data	R21-2	L1, L3, L8	5.54
166	DQ15	Data	R20-2	L1, L3, L8	5.45
168	DQ11	Data	R17-2	L1, L3, L8	5.45

170	DQ21	Data	R36-2	L1, L3, L8	5.45
172	DQ17	Data	R27-2	L1, L3, L8	5.47
174	DQS2_C	Data	R34-2	L1, L3, L8	5.56
175	DQS2_T	Data	R33-2	L1, L3, L8	5.56
177	DQ23	Data	R32-2	L1, L3, L8	5.45
179	DQ19	Data	R29-2	L1, L3, L8	5.45
181	DQ29	Data	R48-2	L1, L3, L8	5.49
183	DQ25	Data	R39-2	L1, L3, L8	5.53
185	DQS3_C	Data	R46-2	L1, L3, L8	5.67
186	DQS3_T	Data	R45-2	L1, L3, L8	5.71
188	DQ31	Data	R44-2	L1, L3, L8	5.64
190	DQ27	Data	R41-2	L1, L3, L8	5.68
207	BG1	Address	R98-1, U1-J8, U2-J8, U3-J8, U4-J8, U5-J8, U6-J8, U7-J8, U8-J8	L1, L3, L5, L6, L8	4.53
208	ALERT_n	ALERT	R102-1, U1-L9, U2-L9, U3-L9, U4-L9, U5-L9, U6-L9, U7-L9, U8-L9	L1, L3, L5, L6, L8	4.53
210	A11	Address	R122-1, U1-N2, U2-N2, U3-N2, U4-N2, U5-N2, U6-N2, U7-N2, U8-N2	L1, L3, L5, L6, L8	4.53
211	A7	Address	R114-1, U1-M8, U2-M8, U3-M8, U4-M8, U5-M8, U6-M8, U7-M8, U8-M8	L1, L3, L5, L6, L8	4.53
213	A5	Address	R110-1, U1-L8, U2-L8, U3-L8, U4-L8, U5-L8, U6-L8, U7-L8, U8-L8	L1, L3, L5, L6, L8	4.53
214	A4	Address	R108-1, U1-K3, U2-K3, U3-K3, U4-K3, U5-K3, U6-K3, U7-K3, U8-K3	L1, L3, L5, L6, L8	4.53
216	A2	Address	R103-1, U1-M3, U2-M3, U3-M3, U4-M3, U5-M3, U6-M3, U7-M3, U8-M3	L1, L3, L5, L6, L8	4.53
218	CK1_T	Clock	R106-1	L1, L6, L8	0.00
219	CK1_C	Clock	R106-2	L1, L6, L8	0.00
222	PAR	Address	R126-1, U1-N3, U2-N3, U3-N3, U4-N3, U5-N3, U6-N3, U7-N3, U8-N3	L1, L3, L5, L6, L8	4.53
224	BA1	Address	R120-1, U1-K8, U2-K8, U3-K8, U4-K8, U5-K8, U6-K8, U7-K8, U8-K8	L1, L3, L5, L6, L8	4.53
225	A10_AP	Address	R121-1, U1-J3, U2-J3, U3-J3, U4-J3, U5-J3, U6-J3, U7-J3, U8-J3	L1, L3, L5, L6, L8	4.53
228	WE	Address	R109-1, U1-H2, U2-H2, U3-H2, U4-H2, U5-H2, U6-H2, U7-H2, U8-H2	L1, L3, L5, L6, L8	4.53
232	A13	Address	R124-1, U1-N8, U2-N8, U3-N8, U4-N8, U5-N8, U6-N8, U7-N8, U8-N8	L1, L3, L5, L6, L8	4.53
238	SA2	SPD	U9-3	any signal layer	0.00
240	DQ37	Data	R60-2	L1, L3, L8	5.70
242	DQ33	Data	R51-2	L1, L3, L8	5.70
244	DQS4_C	Data	R58-2	L1, L3, L8	5.81
245	DQS4_T	Data	R57-2	L1, L3, L8	5.79
247	DQ39	Data	R56-2	L1, L3, L8	5.70
249	DQ35	Data	R53-2	L1, L3, L8	5.70
251	DQ45	Data	R72-2	L1, L3, L8	5.66
253	DQ41	Data	R63-2	L1, L3, L8	5.62
255	DQS5_C	Data	R70-2	L1, L3, L8	5.69
256	DQS5_T	Data	R69-2	L1, L3, L8	5.65
258	DQ47	Data	R68-2	L1, L3, L8	5.51
260	DQ43	Data	R65-2	L1, L3, L8	5.47
262	DQ53	Data	R84-2	L1, L3, L8	5.47
264	DQ49	Data	R75-2	L1, L3, L8	5.45
266	DQS6_C	Data	R82-2	L1, L3, L8	5.54
267	DQS6_T	Data	R81-2	L1, L3, L8	5.54
269	DQ55	Data	R80-2	L1, L3, L8	5.45
271	DQ51	Data	R77-2	L1, L3, L8	5.45

273	DQ61	Data	R96-2	L1, L3, L8	5.45
275	DQ57	Data	R87-2	L1, L3, L8	5.45
277	DQS7_C	Data	R94-2	L1, L3, L8	5.54
278	DQS7_T	Data	R93-2	L1, L3, L8	5.56
280	DQ63	Data	R92-2	L1, L3, L8	5.45
282	DQ59	Data	R89-2	L1, L3, L8	5.45
285	SDA	SPD	U9-5	any signal layer	0.00

3.1 U1 - DRAM

DRAMChip DDR4 SDRAM 16Gbit 2Gx8 1.2V 78-Pin FBGA Tray. Data balls take their own 15 ohm resistor; address, command and clock balls are shared with the other seven DRAMs on the fly-by.

Ball	Net	Class	Goes to	Layers	copper mm
A7	NetR11_1	Data	R11-1	L1, L3, L8	6.00
B3	NetR10_1	Data	R10-1	L1, L3, L8	6.68
B7	NetR3_1	Data	R3-1	L1, L3, L8	4.00
B9	NetR1_1	Data	R1-1	L1, L3, L8	0.57
C2	NetR2_1	Data	R2-1	L1, L3, L8	9.43
C3	NetR9_1	Data	R9-1	L1, L3, L8	7.01
D2	NetR6_1	Data	R6-1	L1, L3, L8	11.02
D3	NetR4_1	Data	R4-1	L1, L3, L8	8.27
D7	NetR5_1	Data	R5-1	L1, L3, L8	11.21
D8	NetR12_1	Data	R12-1	L1, L3, L8	5.92
E3	NetR7_1	Data	R7-1	L1, L3, L8	12.06
E7	NetR8_1	Data	R8-1	L1, L3, L8	13.11
F3	ODT	Control	J1-87, R107-1, U2-F3, U3-F3, U4-F3, U5-F3, U6-F3, U7-F3, U8-F3	L1, L3, L5, L6, L8	4.53
F7	CK0_T	Clock	J1-74, R115-1, U2-F7, U3-F7, U4-F7, U5-F7, U6-F7, U7-F7, U8-F7	L1, L6, L8	4.53
F8	CK0_C	Clock	J1-75, R117-1, U2-F8, U3-F8, U4-F8, U5-F8, U6-F8, U7-F8, U8-F8	L1, L6, L8	4.53
G3	CKE	Control	J1-60, R104-1, U2-G3, U3-G3, U4-G3, U5-G3, U6-G3, U7-G3, U8-G3	L1, L3, L5, L6, L8	4.53
G7	CS	Control	J1-84, R101-1, U2-G7, U3-G7, U4-G7, U5-G7, U6-G7, U7-G7, U8-G7	L1, L3, L5, L6, L8	4.53
H2	WE	Address	J1-228, R109-1, U2-H2, U3-H2, U4-H2, U5-H2, U6-H2, U7-H2, U8-H2	L1, L3, L5, L6, L8	4.53
H3	ACT	Address	J1-62, R125-1, U2-H3, U3-H3, U4-H3, U5-H3, U6-H3, U7-H3, U8-H3	L1, L3, L5, L6, L8	4.53
H7	A15_CAS	Address	J1-86, R111-1, U2-H7, U3-H7, U4-H7, U5-H7, U6-H7, U7-H7, U8-H7	L1, L3, L5, L6, L8	4.53
H8	A16_RAS	Address	J1-82, R113-1, U2-H8, U3-H8, U4-H8, U5-H8, U6-H8, U7-H8, U8-H8	L1, L3, L5, L6, L8	4.53
J2	BG0	Address	J1-63, R99-1, U2-J2, U3-J2, U4-J2, U5-J2, U6-J2, U7-J2, U8-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U2-J3, U3-J3, U4-J3, U5-J3, U6-J3, U7-J3, U8-J3	L1, L3, L5, L6, L8	4.53
J7	A12_BC	Address	J1-65, R123-1, U2-J7, U3-J7, U4-J7, U5-J7, U6-J7, U7-J7, U8-J7	L1, L3, L5, L6, L8	4.53
J8	BG1	Address	J1-207, R98-1, U2-J8, U3-J8, U4-J8, U5-J8, U6-J8, U7-J8, U8-J8	L1, L3, L5, L6, L8	4.53
K2	BA0	Address	J1-81, R118-1, U2-K2, U3-K2, U4-K2, U5-K2, U6-K2, U7-K2, U8-K2	L1, L3, L5, L6, L8	4.53
K3	A4	Address	J1-214, R108-1, U2-K3, U3-K3, U4-K3, U5-K3, U6-K3, U7-K3, U8-K3	L1, L3, L5, L6, L8	4.53
K7	A3	Address	J1-71, R105-1, U2-K7, U3-K7, U4-K7, U5-K7, U6-K7, U7-K7, U8-K7	L1, L3, L5, L6, L8	4.53
K8	BA1	Address	J1-224, R120-1, U2-K8, U3-K8, U4-K8, U5-K8, U6-K8, U7-K8, U8-K8	L1, L3, L5, L6, L8	4.53
L1	RESET_N	Control	J1-58, U2-L1, U3-L1, U4-L1, U5-L1, U6-L1, U7-L1, U8-L1	L1, L3, L5, L6, L8	4.53
L2	A6	Address	J1-69, R112-1, U2-L2, U3-L2, U4-L2, U5-L2, U6-L2, U7-L2, U8-L2	L1, L3, L5, L6, L8	4.53
L3	A0	Address	J1-79, R97-1, U2-L3, U3-L3, U4-L3, U5-L3, U6-L3, U7-L3, U8-L3	L1, L3, L5, L6, L8	4.53
L7	A1	Address	J1-72, R100-1, U2-L7, U3-L7, U4-L7, U5-L7, U6-L7, U7-L7, U8-L7	L1, L3, L5, L6, L8	4.53

L8	A5	Address	J1-213, R110-1, U2-L8, U3-L8, U4-L8, U5-L8, U6-L8, U7-L8, U8-L8	L1, L3, L5, L6, L8	4.53
L9	ALERT_n	ALERT	J1-208, R102-1, U2-L9, U3-L9, U4-L9, U5-L9, U6-L9, U7-L9, U8-L9	L1, L3, L5, L6, L8	4.53
M2	A8	Address	J1-68, R116-1, U2-M2, U3-M2, U4-M2, U5-M2, U6-M2, U7-M2, U8-M2	L1, L3, L5, L6, L8	4.53
M3	A2	Address	J1-216, R103-1, U2-M3, U3-M3, U4-M3, U5-M3, U6-M3, U7-M3, U8-M3	L1, L3, L5, L6, L8	4.53
M7	A9	Address	J1-66, R119-1, U2-M7, U3-M7, U4-M7, U5-M7, U6-M7, U7-M7, U8-M7	L1, L3, L5, L6, L8	4.53
M8	A7	Address	J1-211, R114-1, U2-M8, U3-M8, U4-M8, U5-M8, U6-M8, U7-M8, U8-M8	L1, L3, L5, L6, L8	4.53
N2	A11	Address	J1-210, R122-1, U2-N2, U3-N2, U4-N2, U5-N2, U6-N2, U7-N2, U8-N2	L1, L3, L5, L6, L8	4.53
N3	PAR	Address	J1-222, R126-1, U2-N3, U3-N3, U4-N3, U5-N3, U6-N3, U7-N3, U8-N3	L1, L3, L5, L6, L8	4.53
N8	A13	Address	J1-232, R124-1, U2-N8, U3-N8, U4-N8, U5-N8, U6-N8, U7-N8, U8-N8	L1, L3, L5, L6, L8	4.53

3.2 U2 - DRAM

DRAMChip DDR4 SDRAM 16Gbit 2Gx8 1.2V 78-Pin FBGA Tray. Data balls take their own 15 ohm resistor; address, command and clock balls are shared with the other seven DRAMs on the fly-by.

Ball	Net	Class	Goes to	Layers	copper mm
A7	NetR23_1	Data	R23-1	L1, L3, L8	0.57
B3	NetR22_1	Data	R22-1	L1, L3, L8	8.24
B7	NetR15_1	Data	R15-1	L1, L3, L8	0.74
B9	NetR13_1	Data	R13-1	L1, L3, L8	0.57
C2	NetR14_1	Data	R14-1	L1, L3, L8	0.57
C3	NetR21_1	Data	R21-1	L1, L3, L8	8.24
D2	NetR18_1	Data	R18-1	L1, L3, L8	0.57
D3	NetR16_1	Data	R16-1	L1, L3, L8	0.57
D7	NetR17_1	Data	R17-1	L1, L3, L8	0.57
D8	NetR24_1	Data	R24-1	L1, L3, L8	7.95
E3	NetR19_1	Data	R19-1	L1, L3, L8	0.57
E7	NetR20_1	Data	R20-1	L1, L3, L8	0.57
F3	ODT	Control	J1-87, R107-1, U1-F3, U3-F3, U4-F3, U5-F3, U6-F3, U7-F3, U8-F3	L1, L3, L5, L6, L8	4.53
F7	CK0_T	Clock	J1-74, R115-1, U1-F7, U3-F7, U4-F7, U5-F7, U6-F7, U7-F7, U8-F7	L1, L6, L8	4.53
F8	CK0_C	Clock	J1-75, R117-1, U1-F8, U3-F8, U4-F8, U5-F8, U6-F8, U7-F8, U8-F8	L1, L6, L8	4.53
G3	CKE	Control	J1-60, R104-1, U1-G3, U3-G3, U4-G3, U5-G3, U6-G3, U7-G3, U8-G3	L1, L3, L5, L6, L8	4.53
G7	CS	Control	J1-84, R101-1, U1-G7, U3-G7, U4-G7, U5-G7, U6-G7, U7-G7, U8-G7	L1, L3, L5, L6, L8	4.53
H2	WE	Address	J1-228, R109-1, U1-H2, U3-H2, U4-H2, U5-H2, U6-H2, U7-H2, U8-H2	L1, L3, L5, L6, L8	4.53
H3	ACT	Address	J1-62, R125-1, U1-H3, U3-H3, U4-H3, U5-H3, U6-H3, U7-H3, U8-H3	L1, L3, L5, L6, L8	4.53
H7	A15_CAS	Address	J1-86, R111-1, U1-H7, U3-H7, U4-H7, U5-H7, U6-H7, U7-H7, U8-H7	L1, L3, L5, L6, L8	4.53
H8	A16_RAS	Address	J1-82, R113-1, U1-H8, U3-H8, U4-H8, U5-H8, U6-H8, U7-H8, U8-H8	L1, L3, L5, L6, L8	4.53
J2	BG0	Address	J1-63, R99-1, U1-J2, U3-J2, U4-J2, U5-J2, U6-J2, U7-J2, U8-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U1-J3, U3-J3, U4-J3, U5-J3, U6-J3, U7-J3, U8-J3	L1, L3, L5, L6, L8	4.53
J7	A12_BC	Address	J1-65, R123-1, U1-J7, U3-J7, U4-J7, U5-J7, U6-J7, U7-J7, U8-J7	L1, L3, L5, L6, L8	4.53
J8	BG1	Address	J1-207, R98-1, U1-J8, U3-J8, U4-J8, U5-J8, U6-J8, U7-J8, U8-J8	L1, L3, L5, L6, L8	4.53
K2	BA0	Address	J1-81, R118-1, U1-K2, U3-K2, U4-K2, U5-K2, U6-K2, U7-K2, U8-K2	L1, L3, L5, L6, L8	4.53
K3	A4	Address	J1-214, R108-1, U1-K3, U3-K3, U4-K3, U5-K3, U6-K3, U7-K3, U8-K3	L1, L3, L5, L6, L8	4.53
K7	A3	Address	J1-71, R105-1, U1-K7, U3-K7, U4-K7, U5-K7, U6-K7, U7-K7, U8-K7	L1, L3, L5, L6, L8	4.53
K8	BA1	Address	J1-224, R120-1, U1-K8, U3-K8, U4-K8, U5-K8, U6-K8, U7-K8, U8-K8	L1, L3, L5, L6, L8	4.53
L1	RESET_N	Control	J1-58, U1-L1, U3-L1, U4-L1, U5-L1, U6-L1, U7-L1, U8-L1	L1, L3, L5, L6, L8	4.53
L2	A6	Address	J1-69, R112-1, U1-L2, U3-L2, U4-L2, U5-L2, U6-L2, U7-L2, U8-L2	L1, L3, L5, L6, L8	4.53

L3	A0	Address	J1-79, R97-1, U1-L3, U3-L3, U4-L3, U5-L3, U6-L3, U7-L3, U8-L3	L1, L3, L5, L6, L8	4.53
L7	A1	Address	J1-72, R100-1, U1-L7, U3-L7, U4-L7, U5-L7, U6-L7, U7-L7, U8-L7	L1, L3, L5, L6, L8	4.53
L8	A5	Address	J1-213, R110-1, U1-L8, U3-L8, U4-L8, U5-L8, U6-L8, U7-L8, U8-L8	L1, L3, L5, L6, L8	4.53
L9	ALERT_n	ALERT	J1-208, R102-1, U1-L9, U3-L9, U4-L9, U5-L9, U6-L9, U7-L9, U8-L9	L1, L3, L5, L6, L8	4.53
M2	A8	Address	J1-68, R116-1, U1-M2, U3-M2, U4-M2, U5-M2, U6-M2, U7-M2, U8-M2	L1, L3, L5, L6, L8	4.53
M3	A2	Address	J1-216, R103-1, U1-M3, U3-M3, U4-M3, U5-M3, U6-M3, U7-M3, U8-M3	L1, L3, L5, L6, L8	4.53
M7	A9	Address	J1-66, R119-1, U1-M7, U3-M7, U4-M7, U5-M7, U6-M7, U7-M7, U8-M7	L1, L3, L5, L6, L8	4.53
M8	A7	Address	J1-211, R114-1, U1-M8, U3-M8, U4-M8, U5-M8, U6-M8, U7-M8, U8-M8	L1, L3, L5, L6, L8	4.53
N2	A11	Address	J1-210, R122-1, U1-N2, U3-N2, U4-N2, U5-N2, U6-N2, U7-N2, U8-N2	L1, L3, L5, L6, L8	4.53
N3	PAR	Address	J1-222, R126-1, U1-N3, U3-N3, U4-N3, U5-N3, U6-N3, U7-N3, U8-N3	L1, L3, L5, L6, L8	4.53
N8	A13	Address	J1-232, R124-1, U1-N8, U3-N8, U4-N8, U5-N8, U6-N8, U7-N8, U8-N8	L1, L3, L5, L6, L8	4.53

3.3 U3 - DRAM

DRAMChip DDR4 SDRAM 16Gbit 2Gx8 1.2V 78-Pin FBGA Tray. Data balls take their own 15 ohm resistor; address, command and clock balls are shared with the other seven DRAMs on the fly-by.

Ball	Net	Class	Goes to	Layers	copper mm
A7	NetR35_1	Data	R35-1	L1, L3, L8	0.57
B3	NetR34_1	Data	R34-1	L1, L3, L8	9.87
B7	NetR27_1	Data	R27-1	L1, L3, L8	0.57
B9	NetR25_1	Data	R25-1	L1, L3, L8	0.57
C2	NetR26_1	Data	R26-1	L1, L3, L8	0.57
C3	NetR33_1	Data	R33-1	L1, L3, L8	8.12
D2	NetR30_1	Data	R30-1	L1, L3, L8	0.57
D3	NetR28_1	Data	R28-1	L1, L3, L8	0.57
D7	NetR29_1	Data	R29-1	L1, L3, L8	0.57
D8	NetR36_1	Data	R36-1	L1, L3, L8	0.57
E3	NetR31_1	Data	R31-1	L1, L3, L8	0.57
E7	NetR32_1	Data	R32-1	L1, L3, L8	0.57
F3	ODT	Control	J1-87, R107-1, U1-F3, U2-F3, U4-F3, U5-F3, U6-F3, U7-F3, U8-F3	L1, L3, L5, L6, L8	4.53
F7	CK0_T	Clock	J1-74, R115-1, U1-F7, U2-F7, U4-F7, U5-F7, U6-F7, U7-F7, U8-F7	L1, L6, L8	4.53
F8	CK0_C	Clock	J1-75, R117-1, U1-F8, U2-F8, U4-F8, U5-F8, U6-F8, U7-F8, U8-F8	L1, L6, L8	4.53
G3	CKE	Control	J1-60, R104-1, U1-G3, U2-G3, U4-G3, U5-G3, U6-G3, U7-G3, U8-G3	L1, L3, L5, L6, L8	4.53
G7	CS	Control	J1-84, R101-1, U1-G7, U2-G7, U4-G7, U5-G7, U6-G7, U7-G7, U8-G7	L1, L3, L5, L6, L8	4.53
H2	WE	Address	J1-228, R109-1, U1-H2, U2-H2, U4-H2, U5-H2, U6-H2, U7-H2, U8-H2	L1, L3, L5, L6, L8	4.53
H3	ACT	Address	J1-62, R125-1, U1-H3, U2-H3, U4-H3, U5-H3, U6-H3, U7-H3, U8-H3	L1, L3, L5, L6, L8	4.53
H7	A15_CAS	Address	J1-86, R111-1, U1-H7, U2-H7, U4-H7, U5-H7, U6-H7, U7-H7, U8-H7	L1, L3, L5, L6, L8	4.53
H8	A16_RAS	Address	J1-82, R113-1, U1-H8, U2-H8, U4-H8, U5-H8, U6-H8, U7-H8, U8-H8	L1, L3, L5, L6, L8	4.53
J2	BG0	Address	J1-63, R99-1, U1-J2, U2-J2, U4-J2, U5-J2, U6-J2, U7-J2, U8-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U1-J3, U2-J3, U4-J3, U5-J3, U6-J3, U7-J3, U8-J3	L1, L3, L5, L6, L8	4.53
J7	A12_BC	Address	J1-65, R123-1, U1-J7, U2-J7, U4-J7, U5-J7, U6-J7, U7-J7, U8-J7	L1, L3, L5, L6, L8	4.53
J8	BG1	Address	J1-207, R98-1, U1-J8, U2-J8, U4-J8, U5-J8, U6-J8, U7-J8, U8-J8	L1, L3, L5, L6, L8	4.53
K2	BA0	Address	J1-81, R118-1, U1-K2, U2-K2, U4-K2, U5-K2, U6-K2, U7-K2, U8-K2	L1, L3, L5, L6, L8	4.53
K3	A4	Address	J1-214, R108-1, U1-K3, U2-K3, U4-K3, U5-K3, U6-K3, U7-K3, U8-K3	L1, L3, L5, L6, L8	4.53
K7	A3	Address	J1-71, R105-1, U1-K7, U2-K7, U4-K7, U5-K7, U6-K7, U7-K7, U8-K7	L1, L3, L5, L6, L8	4.53
K8	BA1	Address	J1-224, R120-1, U1-K8, U2-K8, U4-K8, U5-K8, U6-K8, U7-K8, U8-K8	L1, L3, L5, L6, L8	4.53

L1	RESET_N	Control	J1-58, U1-L1, U2-L1, U4-L1, U5-L1, U6-L1, U7-L1, U8-L1	L1, L3, L5, L6, L8	4.53
L2	A6	Address	J1-69, R112-1, U1-L2, U2-L2, U4-L2, U5-L2, U6-L2, U7-L2, U8-L2	L1, L3, L5, L6, L8	4.53
L3	A0	Address	J1-79, R97-1, U1-L3, U2-L3, U4-L3, U5-L3, U6-L3, U7-L3, U8-L3	L1, L3, L5, L6, L8	4.53
L7	A1	Address	J1-72, R100-1, U1-L7, U2-L7, U4-L7, U5-L7, U6-L7, U7-L7, U8-L7	L1, L3, L5, L6, L8	4.53
L8	A5	Address	J1-213, R110-1, U1-L8, U2-L8, U4-L8, U5-L8, U6-L8, U7-L8, U8-L8	L1, L3, L5, L6, L8	4.53
L9	ALERT_n	ALERT	J1-208, R102-1, U1-L9, U2-L9, U4-L9, U5-L9, U6-L9, U7-L9, U8-L9	L1, L3, L5, L6, L8	4.53
M2	A8	Address	J1-68, R116-1, U1-M2, U2-M2, U4-M2, U5-M2, U6-M2, U7-M2, U8-M2	L1, L3, L5, L6, L8	4.53
M3	A2	Address	J1-216, R103-1, U1-M3, U2-M3, U4-M3, U5-M3, U6-M3, U7-M3, U8-M3	L1, L3, L5, L6, L8	4.53
M7	A9	Address	J1-66, R119-1, U1-M7, U2-M7, U4-M7, U5-M7, U6-M7, U7-M7, U8-M7	L1, L3, L5, L6, L8	4.53
M8	A7	Address	J1-211, R114-1, U1-M8, U2-M8, U4-M8, U5-M8, U6-M8, U7-M8, U8-M8	L1, L3, L5, L6, L8	4.53
N2	A11	Address	J1-210, R122-1, U1-N2, U2-N2, U4-N2, U5-N2, U6-N2, U7-N2, U8-N2	L1, L3, L5, L6, L8	4.53
N3	PAR	Address	J1-222, R126-1, U1-N3, U2-N3, U4-N3, U5-N3, U6-N3, U7-N3, U8-N3	L1, L3, L5, L6, L8	4.53
N8	A13	Address	J1-232, R124-1, U1-N8, U2-N8, U4-N8, U5-N8, U6-N8, U7-N8, U8-N8	L1, L3, L5, L6, L8	4.53

3.4 U4 - DRAM

DRAMChip DDR4 SDRAM 16Gbit 2Gx8 1.2V 78-Pin FBGA Tray. Data balls take their own 15 ohm resistor; address, command and clock balls are shared with the other seven DRAMs on the fly-by.

Ball	Net	Class	Goes to	Layers	copper mm
A7	NetR47_1	Data	R47-1	L1, L3, L8	0.57
B3	NetR46_1	Data	R46-1	L1, L3, L8	10.62
B7	NetR39_1	Data	R39-1	L1, L3, L8	0.57
B9	NetR37_1	Data	R37-1	L1, L3, L8	0.57
C2	NetR38_1	Data	R38-1	L1, L3, L8	0.57
C3	NetR45_1	Data	R45-1	L1, L3, L8	10.62
D2	NetR42_1	Data	R42-1	L1, L3, L8	0.57
D3	NetR40_1	Data	R40-1	L1, L3, L8	0.57
D7	NetR41_1	Data	R41-1	L1, L3, L8	0.57
D8	NetR48_1	Data	R48-1	L1, L3, L8	0.57
E3	NetR43_1	Data	R43-1	L1, L3, L8	0.57
E7	NetR44_1	Data	R44-1	L1, L3, L8	0.57
F3	ODT	Control	J1-87, R107-1, U1-F3, U2-F3, U3-F3, U5-F3, U6-F3, U7-F3, U8-F3	L1, L3, L5, L6, L8	4.53
F7	CK0_T	Clock	J1-74, R115-1, U1-F7, U2-F7, U3-F7, U5-F7, U6-F7, U7-F7, U8-F7	L1, L6, L8	4.53
F8	CK0_C	Clock	J1-75, R117-1, U1-F8, U2-F8, U3-F8, U5-F8, U6-F8, U7-F8, U8-F8	L1, L6, L8	4.53
G3	CKE	Control	J1-60, R104-1, U1-G3, U2-G3, U3-G3, U5-G3, U6-G3, U7-G3, U8-G3	L1, L3, L5, L6, L8	4.53
G7	CS	Control	J1-84, R101-1, U1-G7, U2-G7, U3-G7, U5-G7, U6-G7, U7-G7, U8-G7	L1, L3, L5, L6, L8	4.53
H2	WE	Address	J1-228, R109-1, U1-H2, U2-H2, U3-H2, U5-H2, U6-H2, U7-H2, U8-H2	L1, L3, L5, L6, L8	4.53
H3	ACT	Address	J1-62, R125-1, U1-H3, U2-H3, U3-H3, U5-H3, U6-H3, U7-H3, U8-H3	L1, L3, L5, L6, L8	4.53
H7	A15_CAS	Address	J1-86, R111-1, U1-H7, U2-H7, U3-H7, U5-H7, U6-H7, U7-H7, U8-H7	L1, L3, L5, L6, L8	4.53
H8	A16_RAS	Address	J1-82, R113-1, U1-H8, U2-H8, U3-H8, U5-H8, U6-H8, U7-H8, U8-H8	L1, L3, L5, L6, L8	4.53
J2	BG0	Address	J1-63, R99-1, U1-J2, U2-J2, U3-J2, U5-J2, U6-J2, U7-J2, U8-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U1-J3, U2-J3, U3-J3, U5-J3, U6-J3, U7-J3, U8-J3	L1, L3, L5, L6, L8	4.53
J7	A12_BC	Address	J1-65, R123-1, U1-J7, U2-J7, U3-J7, U5-J7, U6-J7, U7-J7, U8-J7	L1, L3, L5, L6, L8	4.53
J8	BG1	Address	J1-207, R98-1, U1-J8, U2-J8, U3-J8, U5-J8, U6-J8, U7-J8, U8-J8	L1, L3, L5, L6, L8	4.53
K2	BA0	Address	J1-81, R118-1, U1-K2, U2-K2, U3-K2, U5-K2, U6-K2, U7-K2, U8-K2	L1, L3, L5, L6, L8	4.53
K3	A4	Address	J1-214, R108-1, U1-K3, U2-K3, U3-K3, U5-K3, U6-K3, U7-K3, U8-K3	L1, L3, L5, L6, L8	4.53

K7	A3	Address	J1-71, R105-1, U1-K7, U2-K7, U3-K7, U5-K7, U6-K7, U7-K7, U8-K7	L1, L3, L5, L6, L8	4.53
K8	BA1	Address	J1-224, R120-1, U1-K8, U2-K8, U3-K8, U5-K8, U6-K8, U7-K8, U8-K8	L1, L3, L5, L6, L8	4.53
L1	RESET_N	Control	J1-58, U1-L1, U2-L1, U3-L1, U5-L1, U6-L1, U7-L1, U8-L1	L1, L3, L5, L6, L8	4.53
L2	A6	Address	J1-69, R112-1, U1-L2, U2-L2, U3-L2, U5-L2, U6-L2, U7-L2, U8-L2	L1, L3, L5, L6, L8	4.53
L3	A0	Address	J1-79, R97-1, U1-L3, U2-L3, U3-L3, U5-L3, U6-L3, U7-L3, U8-L3	L1, L3, L5, L6, L8	4.53
L7	A1	Address	J1-72, R100-1, U1-L7, U2-L7, U3-L7, U5-L7, U6-L7, U7-L7, U8-L7	L1, L3, L5, L6, L8	4.53
L8	A5	Address	J1-213, R110-1, U1-L8, U2-L8, U3-L8, U5-L8, U6-L8, U7-L8, U8-L8	L1, L3, L5, L6, L8	4.53
L9	ALERT_n	ALERT	J1-208, R102-1, U1-L9, U2-L9, U3-L9, U5-L9, U6-L9, U7-L9, U8-L9	L1, L3, L5, L6, L8	4.53
M2	A8	Address	J1-68, R116-1, U1-M2, U2-M2, U3-M2, U5-M2, U6-M2, U7-M2, U8-M2	L1, L3, L5, L6, L8	4.53
M3	A2	Address	J1-216, R103-1, U1-M3, U2-M3, U3-M3, U5-M3, U6-M3, U7-M3, U8-M3	L1, L3, L5, L6, L8	4.53
M7	A9	Address	J1-66, R119-1, U1-M7, U2-M7, U3-M7, U5-M7, U6-M7, U7-M7, U8-M7	L1, L3, L5, L6, L8	4.53
M8	A7	Address	J1-211, R114-1, U1-M8, U2-M8, U3-M8, U5-M8, U6-M8, U7-M8, U8-M8	L1, L3, L5, L6, L8	4.53
N2	A11	Address	J1-210, R122-1, U1-N2, U2-N2, U3-N2, U5-N2, U6-N2, U7-N2, U8-N2	L1, L3, L5, L6, L8	4.53
N3	PAR	Address	J1-222, R126-1, U1-N3, U2-N3, U3-N3, U5-N3, U6-N3, U7-N3, U8-N3	L1, L3, L5, L6, L8	4.53
N8	A13	Address	J1-232, R124-1, U1-N8, U2-N8, U3-N8, U5-N8, U6-N8, U7-N8, U8-N8	L1, L3, L5, L6, L8	4.53

3.5 U5 - DRAM

DRAMChip DDR4 SDRAM 16Gbit 2Gx8 1.2V 78-Pin FBGA Tray. Data balls take their own 15 ohm resistor; address, command and clock balls are shared with the other seven DRAMs on the fly-by.

Ball	Net	Class	Goes to	Layers	copper mm
A7	NetR59_1	Data	R59-1	L1, L3, L8	0.57
B3	NetR58_1	Data	R58-1	L1, L3, L8	7.91
B7	NetR51_1	Data	R51-1	L1, L3, L8	0.57
B9	NetR49_1	Data	R49-1	L1, L3, L8	0.57
C2	NetR50_1	Data	R50-1	L1, L3, L8	0.57
C3	NetR57_1	Data	R57-1	L1, L3, L8	7.91
D2	NetR54_1	Data	R54-1	L1, L3, L8	0.57
D3	NetR52_1	Data	R52-1	L1, L3, L8	0.57
D7	NetR53_1	Data	R53-1	L1, L3, L8	0.57
D8	NetR60_1	Data	R60-1	L1, L3, L8	0.57
E3	NetR55_1	Data	R55-1	L1, L3, L8	0.57
E7	NetR56_1	Data	R56-1	L1, L3, L8	0.57
F3	ODT	Control	J1-87, R107-1, U1-F3, U2-F3, U3-F3, U4-F3, U6-F3, U7-F3, U8-F3	L1, L3, L5, L6, L8	4.53
F7	CK0_T	Clock	J1-74, R115-1, U1-F7, U2-F7, U3-F7, U4-F7, U6-F7, U7-F7, U8-F7	L1, L6, L8	4.53
F8	CK0_C	Clock	J1-75, R117-1, U1-F8, U2-F8, U3-F8, U4-F8, U6-F8, U7-F8, U8-F8	L1, L6, L8	4.53
G3	CKE	Control	J1-60, R104-1, U1-G3, U2-G3, U3-G3, U4-G3, U6-G3, U7-G3, U8-G3	L1, L3, L5, L6, L8	4.53
G7	CS	Control	J1-84, R101-1, U1-G7, U2-G7, U3-G7, U4-G7, U6-G7, U7-G7, U8-G7	L1, L3, L5, L6, L8	4.53
H2	WE	Address	J1-228, R109-1, U1-H2, U2-H2, U3-H2, U4-H2, U6-H2, U7-H2, U8-H2	L1, L3, L5, L6, L8	4.53
H3	ACT	Address	J1-62, R125-1, U1-H3, U2-H3, U3-H3, U4-H3, U6-H3, U7-H3, U8-H3	L1, L3, L5, L6, L8	4.53
H7	A15_CAS	Address	J1-86, R111-1, U1-H7, U2-H7, U3-H7, U4-H7, U6-H7, U7-H7, U8-H7	L1, L3, L5, L6, L8	4.53
H8	A16_RAS	Address	J1-82, R113-1, U1-H8, U2-H8, U3-H8, U4-H8, U6-H8, U7-H8, U8-H8	L1, L3, L5, L6, L8	4.53
J2	BG0	Address	J1-63, R99-1, U1-J2, U2-J2, U3-J2, U4-J2, U6-J2, U7-J2, U8-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U1-J3, U2-J3, U3-J3, U4-J3, U6-J3, U7-J3, U8-J3	L1, L3, L5, L6, L8	4.53
J7	A12_BC	Address	J1-65, R123-1, U1-J7, U2-J7, U3-J7, U4-J7, U6-J7, U7-J7, U8-J7	L1, L3, L5, L6, L8	4.53
J8	BG1	Address	J1-207, R98-1, U1-J8, U2-J8, U3-J8, U4-J8, U6-J8, U7-J8, U8-J8	L1, L3, L5, L6, L8	4.53

K2	BA0	Address	J1-81, R118-1, U1-K2, U2-K2, U3-K2, U4-K2, U6-K2, U7-K2, U8-K2	L1, L3, L5, L6, L8	4.53
K3	A4	Address	J1-214, R108-1, U1-K3, U2-K3, U3-K3, U4-K3, U6-K3, U7-K3, U8-K3	L1, L3, L5, L6, L8	4.53
K7	A3	Address	J1-71, R105-1, U1-K7, U2-K7, U3-K7, U4-K7, U6-K7, U7-K7, U8-K7	L1, L3, L5, L6, L8	4.53
K8	BA1	Address	J1-224, R120-1, U1-K8, U2-K8, U3-K8, U4-K8, U6-K8, U7-K8, U8-K8	L1, L3, L5, L6, L8	4.53
L1	RESET_N	Control	J1-58, U1-L1, U2-L1, U3-L1, U4-L1, U6-L1, U7-L1, U8-L1	L1, L3, L5, L6, L8	4.53
L2	A6	Address	J1-69, R112-1, U1-L2, U2-L2, U3-L2, U4-L2, U6-L2, U7-L2, U8-L2	L1, L3, L5, L6, L8	4.53
L3	A0	Address	J1-79, R97-1, U1-L3, U2-L3, U3-L3, U4-L3, U6-L3, U7-L3, U8-L3	L1, L3, L5, L6, L8	4.53
L7	A1	Address	J1-72, R100-1, U1-L7, U2-L7, U3-L7, U4-L7, U6-L7, U7-L7, U8-L7	L1, L3, L5, L6, L8	4.53
L8	A5	Address	J1-213, R110-1, U1-L8, U2-L8, U3-L8, U4-L8, U6-L8, U7-L8, U8-L8	L1, L3, L5, L6, L8	4.53
L9	ALERT_n	ALERT	J1-208, R102-1, U1-L9, U2-L9, U3-L9, U4-L9, U6-L9, U7-L9, U8-L9	L1, L3, L5, L6, L8	4.53
M2	A8	Address	J1-68, R116-1, U1-M2, U2-M2, U3-M2, U4-M2, U6-M2, U7-M2, U8-M2	L1, L3, L5, L6, L8	4.53
M3	A2	Address	J1-216, R103-1, U1-M3, U2-M3, U3-M3, U4-M3, U6-M3, U7-M3, U8-M3	L1, L3, L5, L6, L8	4.53
M7	A9	Address	J1-66, R119-1, U1-M7, U2-M7, U3-M7, U4-M7, U6-M7, U7-M7, U8-M7	L1, L3, L5, L6, L8	4.53
M8	A7	Address	J1-211, R114-1, U1-M8, U2-M8, U3-M8, U4-M8, U6-M8, U7-M8, U8-M8	L1, L3, L5, L6, L8	4.53
N2	A11	Address	J1-210, R122-1, U1-N2, U2-N2, U3-N2, U4-N2, U6-N2, U7-N2, U8-N2	L1, L3, L5, L6, L8	4.53
N3	PAR	Address	J1-222, R126-1, U1-N3, U2-N3, U3-N3, U4-N3, U6-N3, U7-N3, U8-N3	L1, L3, L5, L6, L8	4.53
N8	A13	Address	J1-232, R124-1, U1-N8, U2-N8, U3-N8, U4-N8, U6-N8, U7-N8, U8-N8	L1, L3, L5, L6, L8	4.53

3.6 U6 - DRAM

DRAMChip DDR4 SDRAM 16Gbit 2Gx8 1.2V 78-Pin FBGA Tray. Data balls take their own 15 ohm resistor; address, command and clock balls are shared with the other seven DRAMs on the fly-by.

Ball	Net	Class	Goes to	Layers	copper mm
A7	NetR71_1	Data	R71-1	L1, L3, L8	0.57
B3	NetR70_1	Data	R70-1	L1, L3, L8	5.96
B7	NetR63_1	Data	R63-1	L1, L3, L8	0.57
B9	NetR61_1	Data	R61-1	L1, L3, L8	0.57
C2	NetR62_1	Data	R62-1	L1, L3, L8	0.57
C3	NetR69_1	Data	R69-1	L1, L3, L8	5.96
D2	NetR66_1	Data	R66-1	L1, L3, L8	0.57
D3	NetR64_1	Data	R64-1	L1, L3, L8	0.57
D7	NetR65_1	Data	R65-1	L1, L3, L8	0.57
D8	NetR72_1	Data	R72-1	L1, L3, L8	0.57
E3	NetR67_1	Data	R67-1	L1, L3, L8	0.57
E7	NetR68_1	Data	R68-1	L1, L3, L8	0.57
F3	ODT	Control	J1-87, R107-1, U1-F3, U2-F3, U3-F3, U4-F3, U5-F3, U7-F3, U8-F3	L1, L3, L5, L6, L8	4.53
F7	CK0_T	Clock	J1-74, R115-1, U1-F7, U2-F7, U3-F7, U4-F7, U5-F7, U7-F7, U8-F7	L1, L6, L8	4.53
F8	CK0_C	Clock	J1-75, R117-1, U1-F8, U2-F8, U3-F8, U4-F8, U5-F8, U7-F8, U8-F8	L1, L6, L8	4.53
G3	CKE	Control	J1-60, R104-1, U1-G3, U2-G3, U3-G3, U4-G3, U5-G3, U7-G3, U8-G3	L1, L3, L5, L6, L8	4.53
G7	CS	Control	J1-84, R101-1, U1-G7, U2-G7, U3-G7, U4-G7, U5-G7, U7-G7, U8-G7	L1, L3, L5, L6, L8	4.53
H2	WE	Address	J1-228, R109-1, U1-H2, U2-H2, U3-H2, U4-H2, U5-H2, U7-H2, U8-H2	L1, L3, L5, L6, L8	4.53
H3	ACT	Address	J1-62, R125-1, U1-H3, U2-H3, U3-H3, U4-H3, U5-H3, U7-H3, U8-H3	L1, L3, L5, L6, L8	4.53
H7	A15_CAS	Address	J1-86, R111-1, U1-H7, U2-H7, U3-H7, U4-H7, U5-H7, U7-H7, U8-H7	L1, L3, L5, L6, L8	4.53
H8	A16_RAS	Address	J1-82, R113-1, U1-H8, U2-H8, U3-H8, U4-H8, U5-H8, U7-H8, U8-H8	L1, L3, L5, L6, L8	4.53
J2	BG0	Address	J1-63, R99-1, U1-J2, U2-J2, U3-J2, U4-J2, U5-J2, U7-J2, U8-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U1-J3, U2-J3, U3-J3, U4-J3, U5-J3, U7-J3, U8-J3	L1, L3, L5, L6, L8	4.53

J2	BG0	Address	J1-63, R99-1, U1-J2, U2-J2, U3-J2, U4-J2, U5-J2, U6-J2, U8-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U1-J3, U2-J3, U3-J3, U4-J3, U5-J3, U6-J3, U8-J3	L1, L3, L5, L6, L8	4.53
J7	A12_BC	Address	J1-65, R123-1, U1-J7, U2-J7, U3-J7, U4-J7, U5-J7, U6-J7, U8-J7	L1, L3, L5, L6, L8	4.53
J8	BG1	Address	J1-207, R98-1, U1-J8, U2-J8, U3-J8, U4-J8, U5-J8, U6-J8, U8-J8	L1, L3, L5, L6, L8	4.53
K2	BA0	Address	J1-81, R118-1, U1-K2, U2-K2, U3-K2, U4-K2, U5-K2, U6-K2, U8-K2	L1, L3, L5, L6, L8	4.53
K3	A4	Address	J1-214, R108-1, U1-K3, U2-K3, U3-K3, U4-K3, U5-K3, U6-K3, U8-K3	L1, L3, L5, L6, L8	4.53
K7	A3	Address	J1-71, R105-1, U1-K7, U2-K7, U3-K7, U4-K7, U5-K7, U6-K7, U8-K7	L1, L3, L5, L6, L8	4.53
K8	BA1	Address	J1-224, R120-1, U1-K8, U2-K8, U3-K8, U4-K8, U5-K8, U6-K8, U8-K8	L1, L3, L5, L6, L8	4.53
L1	RESET_N	Control	J1-58, U1-L1, U2-L1, U3-L1, U4-L1, U5-L1, U6-L1, U8-L1	L1, L3, L5, L6, L8	4.53
L2	A6	Address	J1-69, R112-1, U1-L2, U2-L2, U3-L2, U4-L2, U5-L2, U6-L2, U8-L2	L1, L3, L5, L6, L8	4.53
L3	A0	Address	J1-79, R97-1, U1-L3, U2-L3, U3-L3, U4-L3, U5-L3, U6-L3, U8-L3	L1, L3, L5, L6, L8	4.53
L7	A1	Address	J1-72, R100-1, U1-L7, U2-L7, U3-L7, U4-L7, U5-L7, U6-L7, U8-L7	L1, L3, L5, L6, L8	4.53
L8	A5	Address	J1-213, R110-1, U1-L8, U2-L8, U3-L8, U4-L8, U5-L8, U6-L8, U8-L8	L1, L3, L5, L6, L8	4.53
L9	ALERT_n	ALERT	J1-208, R102-1, U1-L9, U2-L9, U3-L9, U4-L9, U5-L9, U6-L9, U8-L9	L1, L3, L5, L6, L8	4.53
M2	A8	Address	J1-68, R116-1, U1-M2, U2-M2, U3-M2, U4-M2, U5-M2, U6-M2, U8-M2	L1, L3, L5, L6, L8	4.53
M3	A2	Address	J1-216, R103-1, U1-M3, U2-M3, U3-M3, U4-M3, U5-M3, U6-M3, U8-M3	L1, L3, L5, L6, L8	4.53
M7	A9	Address	J1-66, R119-1, U1-M7, U2-M7, U3-M7, U4-M7, U5-M7, U6-M7, U8-M7	L1, L3, L5, L6, L8	4.53
M8	A7	Address	J1-211, R114-1, U1-M8, U2-M8, U3-M8, U4-M8, U5-M8, U6-M8, U8-M8	L1, L3, L5, L6, L8	4.53
N2	A11	Address	J1-210, R122-1, U1-N2, U2-N2, U3-N2, U4-N2, U5-N2, U6-N2, U8-N2	L1, L3, L5, L6, L8	4.53
N3	PAR	Address	J1-222, R126-1, U1-N3, U2-N3, U3-N3, U4-N3, U5-N3, U6-N3, U8-N3	L1, L3, L5, L6, L8	4.53
N8	A13	Address	J1-232, R124-1, U1-N8, U2-N8, U3-N8, U4-N8, U5-N8, U6-N8, U8-N8	L1, L3, L5, L6, L8	4.53

3.8 U8 - DRAM

DRAMChip DDR4 SDRAM 16Gbit 2Gx8 1.2V 78-Pin FBGA Tray. Data balls take their own 15 ohm resistor; address, command and clock balls are shared with the other seven DRAMs on the fly-by.

Ball	Net	Class	Goes to	Layers	copper mm
A7	NetR95_1	Data	R95-1	L1, L3, L8	0.57
B3	NetR94_1	Data	R94-1	L1, L3, L8	6.45
B7	NetR87_1	Data	R87-1	L1, L3, L8	0.57
B9	NetR85_1	Data	R85-1	L1, L3, L8	0.57
C2	NetR86_1	Data	R86-1	L1, L3, L8	0.57
C3	NetR93_1	Data	R93-1	L1, L3, L8	6.45
D2	NetR90_1	Data	R90-1	L1, L3, L8	0.57
D3	NetR88_1	Data	R88-1	L1, L3, L8	0.57
D7	NetR89_1	Data	R89-1	L1, L3, L8	0.57
D8	NetR96_1	Data	R96-1	L1, L3, L8	0.57
E3	NetR91_1	Data	R91-1	L1, L3, L8	0.57
E7	NetR92_1	Data	R92-1	L1, L3, L8	0.57
F3	ODT	Control	J1-87, R107-1, U1-F3, U2-F3, U3-F3, U4-F3, U5-F3, U6-F3, U7-F3	L1, L3, L5, L6, L8	4.53
F7	CK0_T	Clock	J1-74, R115-1, U1-F7, U2-F7, U3-F7, U4-F7, U5-F7, U6-F7, U7-F7	L1, L6, L8	4.53
F8	CK0_C	Clock	J1-75, R117-1, U1-F8, U2-F8, U3-F8, U4-F8, U5-F8, U6-F8, U7-F8	L1, L6, L8	4.53
G3	CKE	Control	J1-60, R104-1, U1-G3, U2-G3, U3-G3, U4-G3, U5-G3, U6-G3, U7-G3	L1, L3, L5, L6, L8	4.53
G7	CS	Control	J1-84, R101-1, U1-G7, U2-G7, U3-G7, U4-G7, U5-G7, U6-G7, U7-G7	L1, L3, L5, L6, L8	4.53
H2	WE	Address	J1-228, R109-1, U1-H2, U2-H2, U3-H2, U4-H2, U5-H2, U6-H2, U7-H2	L1, L3, L5, L6, L8	4.53

H3	ACT	Address	J1-62, R125-1, U1-H3, U2-H3, U3-H3, U4-H3, U5-H3, U6-H3, U7-H3	L1, L3, L5, L6, L8	4.53
H7	A15_CAS	Address	J1-86, R111-1, U1-H7, U2-H7, U3-H7, U4-H7, U5-H7, U6-H7, U7-H7	L1, L3, L5, L6, L8	4.53
H8	A16_RAS	Address	J1-82, R113-1, U1-H8, U2-H8, U3-H8, U4-H8, U5-H8, U6-H8, U7-H8	L1, L3, L5, L6, L8	4.53
J2	BG0	Address	J1-63, R99-1, U1-J2, U2-J2, U3-J2, U4-J2, U5-J2, U6-J2, U7-J2	L1, L3, L5, L6, L8	4.53
J3	A10_AP	Address	J1-225, R121-1, U1-J3, U2-J3, U3-J3, U4-J3, U5-J3, U6-J3, U7-J3	L1, L3, L5, L6, L8	4.53
J7	A12_BC	Address	J1-65, R123-1, U1-J7, U2-J7, U3-J7, U4-J7, U5-J7, U6-J7, U7-J7	L1, L3, L5, L6, L8	4.53
J8	BG1	Address	J1-207, R98-1, U1-J8, U2-J8, U3-J8, U4-J8, U5-J8, U6-J8, U7-J8	L1, L3, L5, L6, L8	4.53
K2	BA0	Address	J1-81, R118-1, U1-K2, U2-K2, U3-K2, U4-K2, U5-K2, U6-K2, U7-K2	L1, L3, L5, L6, L8	4.53
K3	A4	Address	J1-214, R108-1, U1-K3, U2-K3, U3-K3, U4-K3, U5-K3, U6-K3, U7-K3	L1, L3, L5, L6, L8	4.53
K7	A3	Address	J1-71, R105-1, U1-K7, U2-K7, U3-K7, U4-K7, U5-K7, U6-K7, U7-K7	L1, L3, L5, L6, L8	4.53
K8	BA1	Address	J1-224, R120-1, U1-K8, U2-K8, U3-K8, U4-K8, U5-K8, U6-K8, U7-K8	L1, L3, L5, L6, L8	4.53
L1	RESET_N	Control	J1-58, U1-L1, U2-L1, U3-L1, U4-L1, U5-L1, U6-L1, U7-L1	L1, L3, L5, L6, L8	4.53
L2	A6	Address	J1-69, R112-1, U1-L2, U2-L2, U3-L2, U4-L2, U5-L2, U6-L2, U7-L2	L1, L3, L5, L6, L8	4.53
L3	A0	Address	J1-79, R97-1, U1-L3, U2-L3, U3-L3, U4-L3, U5-L3, U6-L3, U7-L3	L1, L3, L5, L6, L8	4.53
L7	A1	Address	J1-72, R100-1, U1-L7, U2-L7, U3-L7, U4-L7, U5-L7, U6-L7, U7-L7	L1, L3, L5, L6, L8	4.53
L8	A5	Address	J1-213, R110-1, U1-L8, U2-L8, U3-L8, U4-L8, U5-L8, U6-L8, U7-L8	L1, L3, L5, L6, L8	4.53
L9	ALERT_n	ALERT	J1-208, R102-1, U1-L9, U2-L9, U3-L9, U4-L9, U5-L9, U6-L9, U7-L9	L1, L3, L5, L6, L8	4.53
M2	A8	Address	J1-68, R116-1, U1-M2, U2-M2, U3-M2, U4-M2, U5-M2, U6-M2, U7-M2	L1, L3, L5, L6, L8	4.53
M3	A2	Address	J1-216, R103-1, U1-M3, U2-M3, U3-M3, U4-M3, U5-M3, U6-M3, U7-M3	L1, L3, L5, L6, L8	4.53
M7	A9	Address	J1-66, R119-1, U1-M7, U2-M7, U3-M7, U4-M7, U5-M7, U6-M7, U7-M7	L1, L3, L5, L6, L8	4.53
M8	A7	Address	J1-211, R114-1, U1-M8, U2-M8, U3-M8, U4-M8, U5-M8, U6-M8, U7-M8	L1, L3, L5, L6, L8	4.53
N2	A11	Address	J1-210, R122-1, U1-N2, U2-N2, U3-N2, U4-N2, U5-N2, U6-N2, U7-N2	L1, L3, L5, L6, L8	4.53
N3	PAR	Address	J1-222, R126-1, U1-N3, U2-N3, U3-N3, U4-N3, U5-N3, U6-N3, U7-N3	L1, L3, L5, L6, L8	4.53
N8	A13	Address	J1-232, R124-1, U1-N8, U2-N8, U3-N8, U4-N8, U5-N8, U6-N8, U7-N8	L1, L3, L5, L6, L8	4.53

4. U9 - SPD EEPROM

No Description Available

Pin	Net	Class	Goes to	Layers	copper mm
1	SA0	SPD	J1-139	any signal layer	0.00
2	SA1	SPD	J1-140	any signal layer	0.00
3	SA2	SPD	J1-238	any signal layer	0.00
5	SDA	SPD	J1-285	any signal layer	0.00
6	SCL	SPD	J1-141	any signal layer	0.00

5. Resistors

Pad 1 and pad 2 of every resistor, with the net on each. The function column says what the part is there for; the layers column applies to the signal side.

Resistor	Part	Pad 1 net	Pad 2 net	Function	Layers	copper mm
R1	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR1_1	GND	240 ohm ZQ to GND	L1	0.57
R2	-	NetR2_1	DQ0	15 ohm series, data	L1, L3, L8	12.38
R3	-	NetR3_1	DQ1	15 ohm series, data	L1, L3, L8	9.45
R4	-	NetR4_1	DQ2	15 ohm series, data	L1, L3, L8	11.22
R5	-	NetR5_1	DQ3	15 ohm series, data	L1, L3, L8	16.66

R6	-	NetR6_1	DQ4	15 ohm series, data	L1, L3, L8	13.97
R7	-	NetR7_1	DQ6	15 ohm series, data	L1, L3, L8	15.01
R8	-	NetR8_1	DQ7	15 ohm series, data	L1, L3, L8	18.56
R9	-	NetR9_1	DQS0_T	15 ohm series, data	L1, L3, L8	12.55
R10	-	NetR10_1	DQS0_C	15 ohm series, data	L1, L3, L8	12.22
R11	-	NetR11_1	DM0	15 ohm series, data	L1, L3, L8	8.95
R12	-	NetR12_1	DQ5	15 ohm series, data	L1, L3, L8	11.37
R13	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR13_1	GND	240 ohm ZQ to GND	L1	0.57
R14	-	NetR14_1	DQ8	15 ohm series, data	L1, L3, L8	3.52
R15	-	NetR15_1	DQ9	15 ohm series, data	L1, L3, L8	6.21
R16	-	NetR16_1	DQ10	15 ohm series, data	L1, L3, L8	3.52
R17	-	NetR17_1	DQ11	15 ohm series, data	L1, L3, L8	6.02
R18	-	NetR18_1	DQ12	15 ohm series, data	L1, L3, L8	3.52
R19	-	NetR19_1	DQ14	15 ohm series, data	L1, L3, L8	3.52
R20	-	NetR20_1	DQ15	15 ohm series, data	L1, L3, L8	6.02
R21	-	NetR21_1	DQS1_T	15 ohm series, data	L1, L3, L8	13.78
R22	-	NetR22_1	DQS1_C	15 ohm series, data	L1, L3, L8	13.78
R23	-	NetR23_1	DM1	15 ohm series, data	L1, L3, L8	3.52
R24	-	NetR24_1	DQ13	15 ohm series, data	L1, L3, L8	13.40
R25	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR25_1	GND	240 ohm ZQ to GND	L1	0.57
R26	-	NetR26_1	DQ16	15 ohm series, data	L1, L3, L8	3.52
R27	-	NetR27_1	DQ17	15 ohm series, data	L1, L3, L8	6.04
R28	-	NetR28_1	DQ18	15 ohm series, data	L1, L3, L8	3.52
R29	-	NetR29_1	DQ19	15 ohm series, data	L1, L3, L8	6.02
R30	-	NetR30_1	DQ20	15 ohm series, data	L1, L3, L8	3.52
R31	-	NetR31_1	DQ22	15 ohm series, data	L1, L3, L8	3.52
R32	-	NetR32_1	DQ23	15 ohm series, data	L1, L3, L8	6.02
R33	-	NetR33_1	DQS2_T	15 ohm series, data	L1, L3, L8	13.69
R34	-	NetR34_1	DQS2_C	15 ohm series, data	L1, L3, L8	15.43
R35	-	NetR35_1	DM2	15 ohm series, data	L1, L3, L8	3.52
R36	-	NetR36_1	DQ21	15 ohm series, data	L1, L3, L8	6.02
R37	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR37_1	GND	240 ohm ZQ to GND	L1	0.57
R38	-	NetR38_1	DQ24	15 ohm series, data	L1, L3, L8	3.58
R39	-	NetR39_1	DQ25	15 ohm series, data	L1, L3, L8	6.10
R40	-	NetR40_1	DQ26	15 ohm series, data	L1, L3, L8	3.72
R41	-	NetR41_1	DQ27	15 ohm series, data	L1, L3, L8	6.24
R42	-	NetR42_1	DQ28	15 ohm series, data	L1, L3, L8	3.54
R43	-	NetR43_1	DQ30	15 ohm series, data	L1, L3, L8	3.68
R44	-	NetR44_1	DQ31	15 ohm series, data	L1, L3, L8	6.20
R45	-	NetR45_1	DQS3_T	15 ohm series, data	L1, L3, L8	16.33
R46	-	NetR46_1	DQS3_C	15 ohm series, data	L1, L3, L8	16.29
R47	-	NetR47_1	DM3	15 ohm series, data	L1, L3, L8	3.62

R48	-	NetR48_1	DQ29	15 ohm series, data	L1, L3, L8	6.06
R49	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR49_1	GND	240 ohm ZQ to GND	L1	0.57
R50	-	NetR50_1	DQ32	15 ohm series, data	L1, L3, L8	3.77
R51	-	NetR51_1	DQ33	15 ohm series, data	L1, L3, L8	6.27
R52	-	NetR52_1	DQ34	15 ohm series, data	L1, L3, L8	3.77
R53	-	NetR53_1	DQ35	15 ohm series, data	L1, L3, L8	6.27
R54	-	NetR54_1	DQ36	15 ohm series, data	L1, L3, L8	3.77
R55	-	NetR55_1	DQ38	15 ohm series, data	L1, L3, L8	3.77
R56	-	NetR56_1	DQ39	15 ohm series, data	L1, L3, L8	6.27
R57	-	NetR57_1	DQS4_T	15 ohm series, data	L1, L3, L8	13.71
R58	-	NetR58_1	DQS4_C	15 ohm series, data	L1, L3, L8	13.73
R59	-	NetR59_1	DM4	15 ohm series, data	L1, L3, L8	3.77
R60	-	NetR60_1	DQ37	15 ohm series, data	L1, L3, L8	6.27
R61	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR61_1	GND	240 ohm ZQ to GND	L1	0.57
R62	-	NetR62_1	DQ40	15 ohm series, data	L1, L3, L8	3.70
R63	-	NetR63_1	DQ41	15 ohm series, data	L1, L3, L8	6.18
R64	-	NetR64_1	DQ42	15 ohm series, data	L1, L3, L8	3.56
R65	-	NetR65_1	DQ43	15 ohm series, data	L1, L3, L8	6.04
R66	-	NetR66_1	DQ44	15 ohm series, data	L1, L3, L8	3.74
R67	-	NetR67_1	DQ46	15 ohm series, data	L1, L3, L8	3.60
R68	-	NetR68_1	DQ47	15 ohm series, data	L1, L3, L8	6.08
R69	-	NetR69_1	DQS5_T	15 ohm series, data	L1, L3, L8	11.61
R70	-	NetR70_1	DQS5_C	15 ohm series, data	L1, L3, L8	11.65
R71	-	NetR71_1	DM5	15 ohm series, data	L1, L3, L8	3.66
R72	-	NetR72_1	DQ45	15 ohm series, data	L1, L3, L8	6.22
R73	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR73_1	GND	240 ohm ZQ to GND	L1	0.57
R74	-	NetR74_1	DQ48	15 ohm series, data	L1, L3, L8	3.52
R75	-	NetR75_1	DQ49	15 ohm series, data	L1, L3, L8	6.02
R76	-	NetR76_1	DQ50	15 ohm series, data	L1, L3, L8	3.52
R77	-	NetR77_1	DQ51	15 ohm series, data	L1, L3, L8	6.02
R78	-	NetR78_1	DQ52	15 ohm series, data	L1, L3, L8	3.52
R79	-	NetR79_1	DQ54	15 ohm series, data	L1, L3, L8	3.52
R80	-	NetR80_1	DQ55	15 ohm series, data	L1, L3, L8	6.02
R81	-	NetR81_1	DQS6_T	15 ohm series, data	L1, L3, L8	11.04
R82	-	NetR82_1	DQS6_C	15 ohm series, data	L1, L3, L8	10.71
R83	-	NetR83_1	DM6	15 ohm series, data	L1, L3, L8	3.52
R84	-	NetR84_1	DQ53	15 ohm series, data	L1, L3, L8	6.04
R85	240R 0.063W 1% 0402 (1005 Metric) SMD	NetR85_1	GND	240 ohm ZQ to GND	L1	0.57
R86	-	NetR86_1	DQ56	15 ohm series, data	L1, L3, L8	3.52
R87	-	NetR87_1	DQ57	15 ohm series, data	L1, L3, L8	6.02
R88	-	NetR88_1	DQ58	15 ohm series, data	L1, L3, L8	3.52

R89	-	NetR89_1	DQ59	15 ohm series, data	L1, L3, L8	6.02
R90	-	NetR90_1	DQ60	15 ohm series, data	L1, L3, L8	3.52
R91	-	NetR91_1	DQ62	15 ohm series, data	L1, L3, L8	3.52
R92	-	NetR92_1	DQ63	15 ohm series, data	L1, L3, L8	6.02
R93	-	NetR93_1	DQS7_T	15 ohm series, data	L1, L3, L8	12.02
R94	-	NetR94_1	DQS7_C	15 ohm series, data	L1, L3, L8	11.99
R95	-	NetR95_1	DM7	15 ohm series, data	L1, L3, L8	3.52
R96	-	NetR96_1	DQ61	15 ohm series, data	L1, L3, L8	6.02
R97	-	A0	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R98	-	BG1	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R99	-	BG0	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R100	-	A1	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R101	-	CS	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R102	-	ALERT_n	VDD	47 ohm ALERT_n pull-up to VDD	L1, L3, L5, L6, L8	4.53
R103	-	A2	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R104	-	CKE	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R105	-	A3	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R106	-	CK1_T	CK1_C	75 ohm across the unused clock pair	L1, L6, L8	0.00
R107	-	ODT	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R108	-	A4	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R109	-	WE	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R110	-	A5	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R111	-	A15_CAS	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R112	-	A6	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R113	-	A16_RAS	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R114	-	A7	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R115	-	CK0_T	NetC56_1	39 ohm clock termination	L1, L6, L8	4.53
R116	-	A8	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R117	-	CK0_C	NetC56_1	39 ohm clock termination	L1, L6, L8	4.53
R118	-	BA0	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R119	-	A9	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53

R120	-	BA1	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R121	-	A10_AP	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R122	-	A11	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R123	-	A12_BC	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R124	-	A13	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R125	-	ACT	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53
R126	-	PAR	VTT	39 ohm termination to VTT	L1, L3, L5, L6, L8	4.53

6. Capacitors

None of these is a routed signal: give each pad its own via straight into the plane it belongs to, as short as you can make it.

Capacitor	Part	Pad 1 net	Pad 2 net	Function
C1	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C2	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C3	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C4	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling
C5	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C6	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C7	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C8	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C9	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C10	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling
C11	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C12	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C13	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C14	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C15	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C16	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling
C17	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C18	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C19	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C20	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C21	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C22	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling
C23	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C24	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C25	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C26	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C27	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C28	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling

C29	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C30	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C31	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C32	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C33	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C34	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling
C35	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C36	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C37	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C38	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C39	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C40	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling
C41	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C42	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C43	CAP CER 0.1UF 16V X7R 0402	VREFCA	VDD	decoupling
C44	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C45	CAP CER 0.1UF 16V X7R 0402	GND	VDD	decoupling
C46	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	VDD	GND	decoupling
C47	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C48	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C49	CAP CER 0.1UF 16V X7R 0402	GND	VDDSPD	decoupling
C50	Multilayer Ceramic Capacitors 4.7µF ±10% 16V X5R SMD 0603	GND	VDD	decoupling
C51	Multilayer Ceramic Capacitors 4.7µF ±10% 16V X5R SMD 0603	GND	VDD	decoupling
C52	Multilayer Ceramic Capacitors 4.7µF ±10% 16V X5R SMD 0603	GND	VDD	decoupling
C53	CAP CER 0.1UF 16V X7R 0402	GND	VPP	decoupling
C54	CAP CER 0.1UF 16V X7R 0402	VDD	VREFCA	decoupling
C55	Multilayer Ceramic Capacitors 4.7µF ±10% 16V X5R SMD 0603	GND	VDD	decoupling
C56	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 10000pF, X7R, 15%, 10%, 50V	NetC56_1	VDD	clock termination node to VDD
C57	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C58	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C59	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C60	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C61	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C62	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C63	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C64	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C65	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C66	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C67	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C68	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C69	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling
C70	CAP CER 0.1UF 16V X7R 0402	VDD	VTT	decoupling